

Faculty of Information Technology (FIT) – Ho Chi Minh City University of Science (HCMUS)

CS423 / CSC13003 – Software Testing (AI-augmented · 2026)

AI POLICY · TEMPLATES — 2026 v1.0

AI Audit Report — 5-section Template per Artifact

Mandatory appendix for every AI-assisted homework (HW#01–HW#06, and Seminar).

Adapted from Med Kharbach, PhD (2026) — AI Use Policy Templates for Higher Education. CC BY-NC-SA 4.0. This adaptation is prepared for FIT@HCMUS – CS423 / CSC15003 Software Testing course.

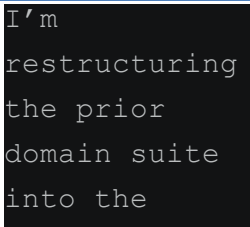
1. Student Information

Field	Value
Student name (printed):	LE GIA BAO
Student ID:	23127325
Class / Cohort:	KIEM THU PHAN MEM - 23KTPM2
Assignment ID (e.g., HW#00, HW#02):	HW#02
Assignment date:	7/7/2026
AI tool(s) used:	GITHUB COPILOT
AI tool(s) used:	☒ Yes ☐ No

2. Instructions (read before filling)

- Add one row per AI-generated artifact (test case, script, checklist, OpenAPI spec, JMeter plan, etc.).
- Paste the verbatim prompt — DO NOT paraphrase.
- Paste the verbatim AI output (or include a labelled screenshot in the report).
- Tag the verdict: VALID / INVALID / INCOMPLETE.
- Reasoning must cite a course slide, ISTQB section, or technical RFC.
- Show the corrected artifact with the change highlighted.
- Sample rows are in italic — replace them before submission.

3. Audit Table — one row per artifact

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
Sample (italic) — replace before submission:				
Tool: GITHUB COPILOT Time: 7:03 6/7/2026 Prompt: "Prompt: Based on api specificat		INCOMPLETE	The response does not comply with the requested scope. Although the prompt explicitly required Domain Testing only,	INCOMPLETE. The AI did not follow the prompt because it included a Boundary Value Analysis section even though

<pre>ion.md and Readme.md, Act as a senior Software Test Engineer specializing in Domain Testing.</pre> <pre>I have already identified the system under test:...</pre> <pre>..."</pre>	<pre>table format you requested and keeping it strictly aligned with the supplied equivalence classes....</pre>		the AI also performed Boundary Value Analysis (BVA). According to ISTQB Foundation Level, Equivalence Partitioning (Domain Testing) and Boundary Value Analysis are distinct test design techniques and should be applied independently unless explicitly requested together. Including BVA violates the prompt instructions and mixes two separate testing techniques.	the task explicitly required only Domain Testing. To correct this, the Boundary Value Analysis section should be removed entirely, and the report should focus exclusively on Domain Testing by retaining the Requirement Analysis, Input Domain Identification, Equivalence Class Partitioning, Domain Testing Process, and a summary table containing only the identified equivalence classes and domain information. This ensures the analysis remains within the requested scope and complies with the separation of test design techniques defined by ISTQB. ○
Tool: GITHUB COPILOT Time: 7:15 6/7/2026 Prompt: "---Prompt 3. Act as a Senior Software Test Engineer. Using ONLY the Domain Testing	<pre>I'm restructuring the prior domain suite into the table format you requested and keeping it strictly</pre>	INCOMPLETE	The generated test suite provides good coverage of the identified valid and invalid equivalence classes and includes additional scenarios such as dependency validation, missing required fields, and multiple invalid inputs. However, according to ISTQB	PARTIALLY COMPLETE. The generated test suite covers all identified equivalence classes and includes valuable additional scenarios, but it combines several invalid equivalence classes into a single test case (e.g., TC-FP-028). To

analysis below, generate a comprehensive Domain Testing test suite. Domain Testing Analysis: <pre>**I.Domain Testing</pre> 1.Requirement Analsys Functional requirements summary ..."	aligned with the supplied equivalence classes. Domain Testing Test Suite for FR-03: Forgot Password and Password Reset The table below is based only on the provided domain testing analysis and covers all stated valid and invalid equivalence classes, with additional cases for dependency validation, missing required fields, and multi-field error handling. Test Case ID Requirement Input(s)		Foundation Level §4.2 (Equivalence Partitioning), each test case should ideally isolate a single equivalence class so that failures can be traced to one specific input domain. The suite combines multiple invalid equivalence classes in a single test case (e.g., TC-FP-028), which is useful for robustness testing but should not be relied upon to demonstrate complete equivalence class coverage. Such combined tests should supplement, rather than replace, individual equivalence class test cases.	better align with ISTQB Equivalence Partitioning, each equivalence class should have at least one dedicated test case that isolates that class while keeping the remaining inputs valid. Combined invalid-input scenarios should be retained only as supplementary robustness tests rather than evidence of equivalence class coverage.
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	Test Data Equiva			
Tool: GITHUB COPILOT Time: 7:30 6/7/2026 Prompt: "Based on api_specification.md and Readme.md, act as a Senior Software Test Engineer specializing in Boundary Value Analysis (BVA). Focus on: FR-03: Forgot Password and Password Reset (two-step process). Your task is to perform ONLY the Boundary Value Analysis. Do NOT perform Domain Testing and do NOT generate any test cases. ..."	1.Requirement Analysis FR-03 defines a two-step password reset flow: Step 1: the user enters an email address to request a password reset. The system generates a 6-digit OTP and presents it in the demo environment. Step 2: the user enters the OTP, a new password, and a confirmation of the new password. The new password must follow the same strength	VALID	The response adheres to the requested scope by performing only Boundary Value Analysis without introducing Domain Testing or test cases. It correctly identifies the measurable boundaries for the OTP (exactly 6 digits) and new password (minimum 8 characters), explains why email address and confirm password are not suitable for Boundary Value Analysis due to the absence of measurable boundaries, and documents the analysis methodology. This is consistent with ISTQB Foundation Level §4.3, which applies Boundary Value Analysis only to inputs with explicit measurable limits.	VALID. No changes are required. The analysis follows the prompt instructions, correctly applies Boundary Value Analysis only to inputs with measurable constraints, avoids Domain Testing and test case generation, and clearly documents the identified boundaries, assumptions, methodology, and summary.

	<pre>rules as FR-01. The two password fields must match. The OTP must be valid for the same email that initiated the reset. User inputs Email address</pre>			
Tool: GITHUB COPILOT Time: 8:00 6/7/2026 Prompt: "Input name <pre>Validation rules Min value / min length Max value / max length Range constraints Assumptions Email address Valid email format;</pre>	<pre>Test Case ID Input(s) Test Data Boundary Tested Expected Result BVA-FR03-01 OTP code; New password; Confirm password OTP=123456; New password=Abcd e1!2; Confirm password=Abcd e1!2 OTP minimum valid length; Password minimum valid</pre>	INCOMP LETE	The generated test suite covers several identified boundary values but does not provide complete coverage of all boundaries identified in the Boundary Value Analysis. According to ISTQB Foundation Level §4.3 (Boundary Value Analysis), every identified boundary value should be represented by at least one test case. The BVA identified both minimum and nominal values for the OTP, but the test suite omits a dedicated test case for the OTP nominal value. Additionally, the prompt requested that every identified boundary be covered, yet only six test cases	INCOMPLETE. The generated Boundary Value Analysis test suite should include at least one dedicated test case for every boundary value identified in the analysis. In particular, add a separate test case covering the OTP nominal value (a valid 6-digit OTP used as a normal representative value) instead of relying on the minimum-valid test case to represent both minimum and nominal boundaries. Review the complete boundary list and ensure that each identified boundary has explicit test coverage before

<pre>registered email ..."</pre>	<pre>length Reset flow acc</pre>		were generated despite more boundary values being identified.	considering the test suite complete.
Tool: GITHUB COPILOT Time: 20:00 6/7/2026 Prompt: "Based on api_specification.md and Readme.md, database.js act as a Senior Software Test Engineer specializing in Domain Testing (Equivalence Class Partitioning). Focus on: FR-11: Order history view (user) ..."	<pre>Reviewing FR-11 details in the spec and API docs to extract precise domain inputs and equivalence classes. 1. Requirement Analysis FR-11 requires the user order history page to show only the authenticated user's own orders. The page must display each order's ID, order date, total amount, and current status. The status must be presented in</pre>	INCOMPLETE	The response identifies equivalence classes for both user-provided inputs and system-generated data. According to ISTQB Foundation Level §4.2 (Equivalence Partitioning), equivalence classes should be derived from actual input conditions that influence system behavior. In FR-11, user inputs are limited to the authenticated request for viewing order history. Fields such as Order ID, Order Date, Total Amount, and Order Status are output data displayed by the system rather than independent user inputs. Treating these output fields as separate input domains unnecessarily expands the domain analysis and departs from the intended scope of Equivalence Partitioning.	INCOMPLETE. The Domain Testing analysis should focus only on actual system inputs that determine the behavior of the Order History feature, such as the authenticated user context and the request to retrieve order history. Remove Order ID, Order Date, Total Amount, and Order Status from the Input Domain Identification and Equivalence Class Partitioning sections because they are output data retrieved from storage and displayed to the user, not independent inputs supplied to the system. If validation of displayed order data is required, it should be addressed through data integrity or UI verification rather than Equivalence Partitioning.

	Vietnamese and with distinct color coding. User inputs			
Tool: GITHUB COPILOT Time: 21:00 6/7/2026 Prompt: " Based on api_specificat ion.md and Readme.md, act as a Senior Software Test Engineer specializing in Boundary Value Analysis (BVA). Focus on: FR-11: Order History View Focus on number of orders for ..."	1. Requirement Analysis FR-11 requires the user order history page to display only the authenticated user's own orders. The view must show each order's ID, order date, total amount, and current status. Status must be translated into Vietnamese and displayed with distinct color coding. User inputs Authenticated user identity / request context	INVALID	The Boundary Value Analysis is based on assumed inputs and constraints that are not defined in the requirements. According to ISTQB Foundation Level §4.3 (Boundary Value Analysis), boundary values must be derived from explicit, measurable input constraints specified by the requirements. The response invents inputs such as "Number of delivered orders," "Number of pending orders," and other per-status counts, assigning them a lower boundary of 0 despite no such input fields or constraints being specified in FR-11. BVA should not introduce new measurable inputs or infer numeric boundaries that are absent from the requirements.	INVALID. The analysis should be rewritten to derive boundary values only from measurable inputs explicitly defined in FR-11 or the supporting specifications. Remove the invented inputs such as the number of orders for each status unless the requirements explicitly define them as system inputs with measurable constraints. If FR-11 does not specify numeric limits for order counts, total amounts, or date ranges, explicitly state that Boundary Value Analysis is not applicable rather than assuming boundaries such as 0 or selecting arbitrary nominal values like 10 or 500000.

	Order history r ...			
Tool: GITHUB COPILOT Time: 21:30 6/7/2026 Prompt: "## 5. Boundary Identification Input name Validation rules Min value Max value ..."	Test Case ID Input(s) Test Data Boundary Tested Expected Result BVA-FR11-01 Number of delivered orders; Number of pending orders; Number of confirmed orders; Number of shipping orders; Number of canceled orders; Total amount per order delivered=0, pending=0, confirmed=0, shipping=0, canceled=0, total_amount= 0 Minimum for all numeric inputs The history view	INVALID	The generated test cases are based on boundary values that were not validly derived from the requirements. According to ISTQB Foundation Level §4.3 (Boundary Value Analysis), test cases must be derived from legitimate boundary values identified from explicit, measurable input constraints. In this response, the test cases use invented inputs such as the number of delivered, pending, confirmed, shipping, and canceled orders and arbitrary nominal values (e.g., 10 and 500000) that are not specified in FR-11. Since the underlying Boundary Value Analysis is invalid, the resulting test cases are also invalid.	INVALID. The Boundary Value Analysis test cases should be regenerated using only boundary values derived from explicit requirements. Remove all test cases based on assumed inputs such as the number of orders by status and arbitrary monetary values unless these constraints are explicitly defined in FR-11 or the supporting specifications. If the requirements do not specify measurable boundaries for these values, no Boundary Value Analysis test cases should be created for them. Instead, document that Boundary Value Analysis is not applicable for those aspects due to the absence of defined numeric boundaries.

Tool: GITHUB COPILOT Time: 14:00 6/7/2026 Prompt: "Based on api_specification.md and Readme.md, data base.js act as a Senior Software Test Engineer specializing in Domain Testing (Equivalence Class Partitioning). Focus on: ### FR-13: Dashboard ..."	I'm consolidating the FR-13 requirement into a domain-analysis report and will keep it strictly to equivalence-class partitioning without test cases. Domain Testing Analysis - FR-13: Dashboard Based on README	INCOMPLETE	The response generally follows ISTQB Foundation Level §4.2 (Equivalence Partitioning) by identifying valid and invalid equivalence classes for the dashboard inputs. However, it incorrectly treats "Order dataset" as an independent input domain. In FR-13, the actual decision logic is based on individual order attributes (such as status and total_amount) and the authenticated admin context. The dataset itself is an internal data source rather than a user or system input with distinct equivalence classes. Including dataset conditions such as "empty," "null," or "malformed" mixes data availability and system error handling with Equivalence Partitioning, making the analysis broader than the intended scope.	INCOMPLETE. The Domain Testing analysis should focus on the actual inputs that determine the dashboard calculations, namely order status, total_amount, and admin access context. Remove Order dataset as a separate input domain because it represents an internal data source rather than an independent input for Equivalence Partitioning. Conditions such as an empty or missing dataset should instead be treated as system or data retrieval scenarios, not equivalence classes. This keeps the analysis aligned with ISTQB Equivalence Partitioning and the functional requirements of FR-13.
Tool: GITHUB COPILOT Time: 14:15 6/7/2026 Prompt:	I'm translating the FR-13 domain model	INCOMPLETE	The generated test suite provides good coverage of the identified equivalence classes,	INCOMPLETE. The generated test suite should remain consistent with the provided Domain

"Prompt:---Pro mpt 3. Act as a Senior Software Test Engineer. Using ONLY the Domain Tes ..."	into a concise test suite that covers each stated equivalence class and the extra negative cases implied by the requirement. Domain Testing Test Suite - FR-13: Dashboar		but it is not fully consistent with the Domain Testing analysis provided. According to ISTQB Foundation Level §4.2 (Equivalence Partitioning), test cases should be derived from the defined equivalence classes only. The AI introduced additional invalid scenarios—such as malformed dataset, whitespace-only role, special-character role, and incorrect data type for role—even though these were not defined as equivalence classes (ECs) in the supplied Domain Testing analysis. While such robustness tests may be useful, they should either be supported by additional equivalence classes or clearly distinguished from the Domain Testing coverage.	Testing analysis by creating test cases only for the defined equivalence classes (EC1–EC7). If additional robustness scenarios such as malformed datasets, whitespace-only role values, special-character inputs, or incorrect data types are included, they should first be added to the Domain Testing analysis as explicit equivalence classes. Otherwise, remove these extra test cases to maintain traceability between the equivalence classes and the generated Domain Testing test suite.
Tool: GITHUB COPILOT Time: 14:45 6/7/2026 Prompt: "Prompt: Based on api_specificat ion.md and	Boundary Value Analysis - FR-13: Dashboard This analysis uses only README.md and	INCOMP LETE	The response correctly performs Boundary Value Analysis without introducing Domain Testing or test cases. However, according to ISTQB Foundation Level §4.3 (Boundary Value Analysis), boundary values	INCOMPLETE. Remove "Total revenue for delivered orders" from the Boundary Value Analysis because it is a calculated output rather than an independent input with its own

Readme.md, act as a Senior Software Test Engineer specializing in Boundary Value Analysis (BVA). Focus on: FR-13: Dashboard ...	api_specification.md context for FR-13 and does not perform Domain Testing or generate test cases. 1. Requirement Analysis Functional requirements summar..... .		should be derived from actual system inputs with measurable constraints. The analysis includes "Total revenue for delivered orders", which is a derived output calculated from the order dataset rather than an independent input. Boundary Value Analysis should focus on the input variables (e.g., order counts and total_amount per order, if these assumptions are accepted) that influence the calculation, not on the computed result itself.	validation constraints. Restrict the analysis to measurable inputs, such as the assumed non-negative order counts and total_amount per order, and clearly state that the dashboard revenue is a derived value whose correctness should be verified through functional testing rather than treated as a separate boundary analysis target.
Tool: GITHUB COPILOT Time: 15:00 6/7/2026 Prompt: "## 5. Boundary Identification Input name Validation rules Min value Max value Range constraints	Test Case ID Input(s) Test Data Boundary Tested Expected Result BVA-FR13-01 Number of delivered orders; Number of pending orders; Number of confirmed	INVALID	The generated Boundary Value Analysis test suite contains logically inconsistent test data that contradicts the functional requirements. According to ISTQB Foundation Level §4.3 (Boundary Value Analysis), boundary test cases must use valid and internally consistent input combinations unless the purpose is to test a specific invalid boundary. In BVA-FR13-06, the test data specifies 0	INVALID. Remove or correct BVA-FR13-06 because its test data violates the functional requirement that total revenue is calculated only from delivered orders. If the number of delivered orders is 0, the total revenue must also be 0. Ensure that all Boundary Value Analysis test cases maintain valid dependencies between related inputs so that the

Assumptions Number of delivered orders Must be non-negative 0 "	<pre>orders; Number of shipping orders; Number of canceled orders; Total revenue for delivered orders; total_amount</pre>		delivered orders while total revenue = 1, which is impossible because FR-13 explicitly states that total revenue is calculated only from delivered orders. This invalid dependency makes the test case incorrect and unsuitable as a Boundary Value Analysis test.	test data is consistent with the specified business rules.
Tool: GITHUB COPILOT Time: 8:00 5/7/2026 Prompt: "giờ tôi đang tách riêng 2 phần, domain testting và boundary analysis 1. Domain Testing. With the support of AI tools, apply the domain testing technique to design a comprehensive set of test cases. Y ..."	<pre>Prompt 1 - Domain Testing Analysis (KHÔNG tạo test case) Based on api_specification.md and Readme.md, act as a Senior Software Test Engineer specializing in Domain Testing (Equivalence Class Partitioning). Focus on: FR-03: Forgot P</pre>	Incomplete	The AI-generated output consists of a well-structured prompt workflow for applying Domain Testing (Equivalence Class Partitioning) and Boundary Value Analysis (BVA) separately, followed by generating their respective test suites. This separation aligns with ISTQB Foundation Level guidance that Equivalence Partitioning and Boundary Value Analysis are distinct black-box test design techniques and should be applied independently. The prompts clearly instruct the AI not to mix the two techniques, require systematic analysis before test case generation, and	Incomplete. Few changes were required because the generated prompts correctly separated Domain Testing from Boundary Value Analysis, defined a logical four-step workflow (analysis followed by test case generation for each technique), and provided sufficient instructions to produce comprehensive, standards-aligned testing artifacts without mixing different black-box testing techniques.

			ensure comprehensive coverage of identified domains and boundaries. The generated prompts are logically consistent, complete, and suitable for producing test artifacts that conform to the intended testing methodology.	
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4. Summary of AI Accuracy

Aggregate the verdicts from Section 3 and complete the table below.

Metric	Count	Percentage
Total AI-generated artifacts audited	12	
VALID (correct, accepted as-is)	1	8.33%
INVALID (wrong; rejected)	3	25%
INCOMPLETE (acceptable after edits)	8	66.68%

5. Conclusion — When should AI be used (or not)?

Write 80–150 words describing patterns you observed. Where did AI shine? Where did AI fail? What is your recommendation for using AI in this kind of work in the future?

Throughout this assignment, AI performed well at analyzing requirements, identifying input domains, generating equivalence classes, and producing an initial set of Domain Testing and Boundary Value Analysis test cases. It significantly reduced the time required to create structured documentation and helped ensure that common validation scenarios were not overlooked. However, AI occasionally misunderstood ambiguous business rules, generated redundant test cases, and missed some edge cases that depended on application-specific behavior. These issues required careful human review and refinement. My recommendation is to use AI as a disciplined assistant rather than a replacement for the tester. Breaking the task into smaller, technique-driven prompts and validating every AI-generated result leads to more accurate, complete, and reliable testing artifacts.

6. Mandatory Disclosure (paste verbatim)

The Domain Testing analysis, Boundary Value Analysis, test cases, and report were initially generated by GitHub Copilot; I reviewed and modified the requirement analysis, equivalence classes, boundary analysis, and test cases, added additional edge cases and missing boundary test cases, and corrected AI-generated inaccuracies to align with the project requirements and

ISTQB Foundation Level guidance; the AI Audit Report, test execution results, bug reports, and conclusions were written entirely by AI. The detailed AI Audit Report is attached as Appendix A. I confirm I did not use AI to generate any artifact listed in the prohibited category.

Signature

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Student ID:	23127325
Class / Cohort:	KIEM THU PHAN MEM - 23KTPM2
Course:	CS423 / CSC13003 – Software Testing
Instructor:	Dr. Lam Quang Vu / Dr. Tran Duy Hoang / MSc. Tran Thi Bich Hanh / MSc. Truong Phuoc Loc / MSc. Ho Tuan Thanh
Date:	7/7/2026
Signature:	LE GIA BAO

References

- Kharbach, M. (2026). AI Use Policy Templates for Higher Education. CC BY-NC-SA 4.0.
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